   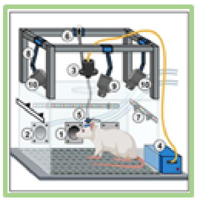 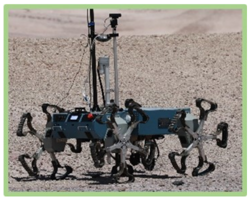

Trevor R. Smith

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Google Scholar: `https://scholar.google.com/citations?user=cfqfknwAAAAJ&hl=en`
Personal Website: `https://trevorsmith42.github.io/`
LinkedIn: `https://www.linkedin.com/in/trevor-smith-05173b147/`

General Information

I research the scalable co-emergence of collective intelligence, morphology, and behavior to form a single coherent machine. I have developed self-organizing modular swarm systems—from multicellular and multi-arm robots to intuitive and adaptive human–machine interfaces for heavy equipment. Across my Ph.D. and current MIT postdoctoral research, I have designed 20+ platforms comprising 200+ physical robots, published 7 first-author papers, and received 4 presentation awards.

Research Interests: Swarm and modular robotics; self-organization; collective and morphological intelligence; decentralized control; adaptive morphology

Applications: Field, construction, agricultural, and space robotics

Education

Aug. 2026 — **Ph.D., Mechanical Engineering**, West Virginia University (WVU), Morgantown, WV
National Science Foundation Graduate Research Fellow
Dissertation: *Swarm of One: Self-Organization in Multicellular Robots*
Advisor: Yu Gu
Committee: Y. Gu, N. Szczecinski, S. Yakovenko, A. Halasz, A. Mishra

Dec. 2021 — **B.S., Mechanical Engineering**, WVU, Morgantown, WV
B.S., Aerospace Engineering
Minor: Computer Science | Summa Cum Laude (4.0 GPA)

Fellowships

2025–2026 — **NASA West Virginia Space Grant Consortium Graduate Research Fellowship**
Competitive one-year state fellowship supporting NASA mission-aligned robotics research.

2022–2025 — **National Science Foundation Graduate Research Fellowship**
Prestigious three-year national fellowship supporting independent doctoral research in STEM disciplines.

Grants Contributed To

2024–2026 — **NSF FRR EAGER: Towards Bottom-Up Problem-Solving with Loopy, A Multicellular Robot**
Funding: $234,544 | Award No. 2422270
Assisted in drafting proposal sections on *Technical Approach*, *Preliminary Work*, *Intellectual Merit*, and *Broader Impacts*.

2021–2026 — **USDA/NIFA NRI: Stickbug — An Effective Co-Robot for Precision Pollination**
Funding: $750,361 | Award No. 2022-67021-36124
Supported proposal preparation by contributing text and figures for the *Technical Approach* and *Preliminary Work* sections.

Awards and Achievements

Apr. 2026 **Outstanding Graduate Student Research Award**
Recognized for demonstrating exemplary academic achievement and research excellence.

WVU Rapid Fire Presentation, 1st Place
Recognized for effectively communicating Ph.D. research to a general engineering audience.

Apr. 2025 **WVU 3-Minute Thesis Competition, 2nd Place**
Recognized for effectively communicating Ph.D. research to a general audience.

July 2024 **Living Machines Conference, Best Presentation Award**
International biomimetics and biohybrid robotics conference; awarded for excellence in research communication.

2023–2025 **University Rover Challenge — Graduate Mentor**
Team placed 1st (2023) and 2nd (2024, 2025) among 100+ international competitors.

Dec. 2021 **WVU Undergraduate Research Symposium Winner — Science & Technology Category**

2017–2021 **WVU Undergraduate Honors and Research Scholarships**
Includes Statler Research, Robotics, and Everette C. Dubbe Scholarships; WVU Honors College graduate.

Research Experience and Publications

Cross-Collaborations: † University of Texas at Dallas Department of Bioengineering, ‡ West Virginia University Department of Neuroscience, § University of Florida Department of Industrial and Systems Engineering

Aug. 2026–Present **Postdoctoral Associate**, MIT, Cambridge, MA
Developing intuitive human–robot interfaces for controlling high-DoF heavy machinery, combining haptic feedback, human motion and intent, and robotic control. Current work investigates naturalistic control of excavators with tiltrotators as a platform for human-centered teleoperation and shared autonomy.

Aug. 2022–2026 **Graduate Research Assistant**, WVU, Morgantown, WV
Research in swarm and bio-inspired robotics focused on self-organization, mechanical intelligence, and adaptive morphology. Developed the *Loopy* multicellular robot to study morphogenesis-driven collective behavior. Extended this work to the six-armed Stickbug co-robot for precision pollination, demonstrating scalable mechanical intelligence. Applied work to distributed automation systems for rodent behavioral neuroscience, integrating real-time sensing, actuation, and feedback.

May 2019–2022 **Undergraduate Research Assistant**, WVU, Morgantown, WV
Investigated multi-agent robotics and developed an experimental testbed for tabletop swarms integrating motion capture, video projection, and low-friction environments. Participated in an NSF Research Experience for Undergraduates (REU) on flocking-based localization using adaptive Boids rules for cooperative positioning.

Journal Articles

2025 †‡ **Smith, T.R.**, Smith, T.J., Szczecinski, N.S., Yakovenko, S., Gu, Y., *Cellular plasticity model for self-organized phenotypes in multi-cellular robots*. npj Robotics
https://doi.org/10.1038/s44182-025-00039-y

2024 † Smith, T. J., **Smith, T. R.**, Faruk, F., Bendea, M., Kumara, S. T., Capadona, J. R., Hernandez-Reynoso, A. & Pancrazio, J. J., *Real-Time Assessment of Rodent Engagement Using ArUco Markers: A Scalable and Accessible Approach for Scoring Behavior in a Nose-Poking Go/No-Go Task*. Eneuro
https://doi.org/10.1523/ENEURO.0500-23.2024

† Smith, T. J., **Smith, T. R.**, Faruk, F., Bendea, M., Kumara, S. T., Capadona, J. R., Hernandez-Reynoso, A.& Pancrazio, J. J., *A Real-Time Approach for Assessing Rodent Engagement in a Nose-Poking Go/No-Go Behavioral Task Using ArUco Markers*. Bio-protocol
https://doi.org/10.21769/BioProtoc.5098

2023 § **Smith, T.**, Chen, Y., Hewitt, N., Hu, B., Gu, Y., *Socially Aware Robot Obstacle Avoidance Considering Human Intention and Preferences*. International Journal of Social Robotics
https://doi.org/10.1007/s12369-021-00795-5

Refereed Conference Proceedings

2026 Rijal, M., Shhrestha, R., **Smith, T.**, & Gu, Y. *Modeling Branches for Active Manipulation using Iterative Parameter Estimation*. International Conference on Intelligent Robots and Systems (IROS)

2025 Shhrestha, R., Rijal, M., **Smith, T.**, & Gu, Y. *FloPE: Flower Pose Estimation for Precision Pollination*. International Conference on Intelligent Robots and Systems (IROS)
https://doi.org/10.1109/IROS60139.2025.11247634

Rijal, M., Shrestha, R., **Smith, T.**, & Gu, Y. *Force-Aware Branch Manipulation To Assist Agricultural Tasks*. International Conference on Intelligent Robots and Systems (IROS)
https://doi.org/10.1109/IROS60139.2025.11246611

† Smith, T. J., Tahmasebi, G., **Smith, T. R.**, Solis, E., Haghighi, P., Pancrazio, J.J., *Modular Tracking System for Treadmill-Based Rodent Experiments*. 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)
https://doi.org/10.1109/EMBC58623.2025.11254127

Smith, T. and Gu, Y. *Loopy Movements: Emergence of Rotation in a Multicellular Robot*, 2025 IEEE International Conference on Robotics and Automation (ICRA)
https://doi.org/10.1109/ICRA55743.2025.11128053

2024 **Smith, T.**, Rijal, M., Tatsch, C., Butts, R. M., Beard, J., Cook, R. T, Chu, A., Gross, J., Gu, Y., *Design of Stickbug: a Six-Armed Precision Pollination Robot*, 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
https://doi.org/10.1109/IROS58592.2024.10801406

†‡ **Smith, T.R.**, Smith, T.J., Szczecinski, N.S., Yakovenko, S., Gu, Y. *Cellular Plasticity Model for Bottom-Up Robotic Design*. In Biomimetic and Biohybrid Systems, Living Machines
https://doi.org/10.1007/978-3-031-72597-5_18

2023 **Smith, T.**, Butts, R. M., Adkins, N., & Gu, Y. *Swarm of One: Bottom-Up Emergence of Stable Robot Bodies from Identical Cells*. In 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
https://doi.org/10.1109/IROS55552.2023.10342118

2022 **Smith, T.**, Gutierrez, E., Bredu, J., Gu, Y., Gross, J., *Cooperative and Coordinated Localization of Swarm Robots using Adaptive Boids Rules*, Proceedings of the 35th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+)
https://doi.org/10.33012/2022.18560

2021 § Chen, Y., **Smith, T.**, Hewitt, N., Gu, Y., & Hu, B. *Effects of Human Personal Space on the Robot Obstacle Avoidance Behavior: A Human-in-the-loop Assessment*. Proceedings of the Human Factors and Ergonomics Society Annual Meeting
`https://doi.org/10.1177/1071181321651098`

Datasets

2024 Rijal, M., **Smith, T.**, Tatsch, C., beard, J., Chu, A., Butts, M., Waterland, N., Gross, J., Gu. Y., *Bramble Flower Detection and Classification Dataset for Precision Pollination*, IEEE Dataport
`https://dx.doi.org/10.21227/b58f-jt53`

Presentations and Demonstrations

2025
- **NSF–FRR/NRI Annual Meeting**, Alexandria, VA, USA.
  Demonstrated *Loopy: A Multicellular Robot for Bottom-Up Problem Solving*.
- **IEEE International Conference on Robotics and Automation (ICRA)**, Atlanta, GA, USA.
  Presented and demonstrated *Loopy Movements: Emergence of Rotation in a Multicellular Robot*.
- **WVU 3-Minute Thesis Competition**. *2nd Place Winner*.

2024
- **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, Abu Dhabi, UAE.
  Presented *Stickbug: A Six-Armed Precision Pollination Robot*.
- **Living Machines Conference**, Chicago, IL, USA.
  Presented *Cellular Plasticity Model for Bottom-Up Robotic Design*. **(Best Presentation Award)**
- **NSF–FRR/NRI Annual Meeting**, Baltimore, MD, USA.
  Demonstrated *Stickbug: Co-Robot for Precision Pollination*.

2023
- **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, Detroit, MI, USA.
  Presented and demonstrated *Swarm of One: Bottom-Up Emergence of Stable Robot Bodies from Identical Cells*.

2022
- **ION GNSS+**, Denver, CO, USA.
  Presented *Cooperative and Coordinated Localization of Swarm Robots Using Adaptive Boids Rules*.

2021
- **WVU Fall Undergraduate Research Symposium**, Morgantown, WV, USA.
  Presented *Gemini Telepresence Robot System Design*. **(Best Presentation Award)**

Media Coverage

Research featured in national and international outlets including CBS, Scientific American, ASME, IEEE Spectrum, The Conversation, and Fast Company.

- Feb. 2026 — **CBS: Visioneers With Zay Harding** — Season 2 Episode 13: *"Machines With Purpose"*
- Jul. 2024 — **ASME** — *“Six-Armed Robot Stickbug Works as a Precision Pollinator.”*
- Feb. 2024 — **The Conversation** — *“Swarm of One Robot Is a Single Machine Made Up of Independent Modules.”*

Jul. 2023 — **Scientific American** — *"Robotic Bees Could Support Vertical Farms Today and Astronauts Tomorrow."*

2021–2023 — **IEEE Spectrum Robotics Blog, Fast Company, and WVU Today** — Coverage of Loopy and Stickbug projects in multiple issues of IEEE's "Video Friday" series and WVU research highlights.

Teaching

Hands-on experience, freedom to explore, and the freedom to make mistakes are critical to learning and innovation in robotics. By equipping students with technical expertise and problem-solving skills, I inspire future roboticists and prepare them for success.

Courses

2025 — **WVU, MAE 412: Mobile Robotics - Design of Education Robot Platform**
Designed, fabricated, and programmed 20 state-of-the-art mobile and manipulator robots for educating students on navigation, sensor fusion, and kinematics.

Summer 2024 — **WVU, MAE 460: Automatic Controls - Guest Lecturer**

Student Mentorship

2023-2026 — **Undergraduate Research Mentorship (9 students, Mechanical, Aerospace, Electrical, Computer Science, and Robotics)**
Mentored students whose projects resulted in co-authorship in peer-reviewed publications (IROS 2024), leadership positions in the University Rover Challenge (CEO, CTO, President), and graduate placements at WVU and in industry (Starlink, Blue Origin, MITRE).

2022 — **NSF Research Experience for Undergraduates (Graduate Mentor)**
Trained 10+ students during a 10-week intensive program to become independent researchers through guided projects in swarm robotics, experimental design, and scientific communication.

2019–Present — **University Rover Challenge Team Mountaineers — Graduate Mentor and Former CTO**
Directed a 40+ member interdisciplinary team in system design, electronics, and software integration. Under mentorship, the team achieved 1st (2023) and 2nd (2024 & 2025) placements among 100+ international teams.

WVU Robotics Club — Graduate Mentor and Former President
Expanded the club from 6 to 30 active members and established a dedicated lab space. Guided students in mechanical design, manufacturing, and programming; the team qualified for the VEX-U World Championship annually (2021–2025).

Professional Service

Memberships

2021–Present — **Institute of Electrical and Electronics Engineers (IEEE)**

Reviewer

2022–Present **Reviewed 20+ submissions for leading robotics venues including:**
• *IEEE Robotics and Automation Letters* (11)
• *International Journal of Social Robotics* (6)
• *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (9)
• *IEEE International Conference on Robotics and Automation (ICRA)* (3)

Outreach

2019–2025 **VEX Robotics Competition (VRC) — Volunteer**
Roles: Scorekeeper and Judge across multiple high school and middle school events per year. Supported *50+ students per event* through field operations, scoring, match coordination, and informal STEM instruction that encouraged teamwork, design thinking, and interest in robotics.

K-12 Lab Visitations and Demonstrations
Inspired and introduced K-12 students to the field of robotics with hands-on robotics demonstrations to an average group size of 10-20 students, 5 times a year.